Assignment - 15

Part - A

1. Maximum modulus theorem

If a function $f(z)$ is analytic and non-constant inside and on a closed bounded region R, then the maximum value of $|f(z)|$ occurs only on the boundary of the region and not at any interior point.

2. Define non-isolated singularity of a complex function.

A singular point $z=a$ of a complex fn. is called a non-isolated singularity if there are infinitely many singular points arbitarily close to a. (No neighbourhood around a exists that contains only one singularity)

3. Isolated singularity and simple pole:

① Isolated singularity:

A point $z=a$ is called an isolated singularity of $f(z)$ if $f(z)$ is not analytic in some neighbourhood around a except at a itself.

② Simple Pole: A pole of order 1 is called a simple pole if

$\lim_{z \to a} (z-a) f(z)$

exists and is finite non-zero, then $z=a$ is a simple pole.

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4. Liouville's Theorem

If a function $f(z)$ is entire (analytic everywhere in the complex plane) and bounded, then $f(z)$ must be constant.

5. Find the zeros of $f(z) = (z+i)^2(z-i)^2 = 0$

$(z+i)^2 = 0$; $(z-i)^2 = 0$

$z = -i$; $z = i$

order : 2 ; : 2

Part - B

1. Evaluate $\oint \frac{3z^2+z^2}{z^2-1} dz$, where C is the circle $|z-1| = 1$

Checking singularity

deno = 0

$(z+1)(z-1) = 0 \Rightarrow \underline{z = 1, -1}$

Check Contour : $|z-1| = 1$ is centered $(1,0)$ with $\underline{\text{rad} = 1}$

$z = 1$ lies inside the circle

$z = -1$ lies outside the circle

$|-1-1| = 2 > 1$

Apply Cauchy's ~~theorem~~ Integral formula:

$g(z) = \frac{3z^2+z}{z+1}$. The integral is $\oint \frac{g(z)}{z-1} dz$

Formula $2\pi i \, g(1)$

$g(1) = \frac{3(1)^2+1}{1+1} = 2$

Result $= 2\pi i \times 2 = \underline{4\pi i}$

Expand $f(z) = \cos z$ in Taylor's Series about $z = \pi/4$

Taylor Series formula is $f(z) = \sum_{n=0}^{\infty} \frac{f^n(a)}{n!} (z-a)^n$

$\rightarrow f(\pi/4) = \cos(\pi/4) = 1/\sqrt{2}$ | $f'''(\pi/4) = -\cos(\pi/4) = -1/\sqrt{2}$

$\rightarrow f'(\pi/4) = -\sin(\pi/4) = -\frac{1}{\sqrt{2}}$ | $f''''(\pi/4) = \sin\frac{\pi}{4} = 1/\sqrt{2}$

Expansion:

$$f(z) = \frac{1}{\sqrt{2}}\left[1 - \left(z - \frac{\pi}{4}\right) - \frac{(z-\pi/4)^2}{2!} + \frac{(z-\pi/4)^3}{3!} - - - \right]$$

3. Condition of Liouville's theorem for $f(z) = 10 - 3i$

(1) Liouville's theorem: An entire function that is bounded must be constant

(2) Check $f(z)$: The function $f(z) = 10 - 3i$ is constant

(3) It is ~~ent~~ extreme and bounded ($|f(z)| = \sqrt{109}$ fixed value)

Part - C

1. Determine Poles and Residues of the following Fn.

$$\oint_0^{2\pi} \frac{1-2z}{z(z-1)(z-2)} \qquad |z| = 1.5$$

$z(z-1)(z-2) = 0 \Rightarrow z = 0, 1, 2$

All are simple poles

$\Rightarrow$ Check Contour $|z| = 1.5$

The circle is centered at the origin with radius 1.5.

→ $z=0$ is inside the contour $(0 < 1.5)$

→ $z=1$ is inside the contour $(1 < 1.5)$

→ $z=2$ is ~~inside~~ outside the contour $(2 > 1.5)$

Calculate Residue for inside poles:

$$\text{Res}(f, a) = \lim_{z \to a} (z-a) f(z)$$

@ $z=0$

$$\text{Res}(f, 0) = \lim_{z \to 0} \frac{1-2z}{(z-1)(z-2)} = \frac{1-0}{(-1)(-2)} = \frac{1}{2}$$

@ $z=1$

$$\text{Res}(f, 1) = \lim_{z \to 1} \frac{1-2z}{z(z-2)} = \frac{-1}{-1} = 1$$

2. Evaluate $\frac{1}{2\pi i} \oint \frac{z^2}{z^2+4} dz$, where C is the vertices at $+2, -2, +4i, -4i$.

Identify singularities

The denominator:

$$x^2 + 4 = 0 \Rightarrow \underline{x = \pm 2i}$$

→ Check Contour

The square was vertices $(2,0), (-2,0), (0,4), (0,-4)$

The point $z = 2i$ and $-2i$ lie on the imaginary axis at $y=2$ and $y=-2$ since the square extends to $y = \pm 4$ on the imaginary axis, both poles are inside the contour.

Calculate Residues

$$f(z) = \frac{z^2}{(z-2i)(z+2i)}$$

ⓐ $z = 2i$; $\text{Res}(f, 2i) = \frac{(2i)^2}{2i+2i} = \frac{-4}{4i} = \frac{-1}{i} = \underline{i}$

ⓐ $z = -2i$; $\text{Res}(f, -2i) = \frac{(-2i)^2}{-2i-2i} = \frac{-4}{-4i} = \frac{1}{i} = \underline{-i}$

⇒ Apply Cauchy residue theorem

$$\oint_C f(z)\,dz = 2\pi i \sum \text{Res}$$

$$= \frac{1}{2\pi i}\left[2\pi i\,(i + (-i))\right] = 0$$

$$\boxed{\therefore \frac{1}{2\pi i}\oint_C \frac{z^2}{z^2+4}\,dz = 0}$$

3 Evaluate Real Integral using complex integration

Evaluate : $I = \int_0^{2\pi} \frac{d\theta}{2+\cos\theta}$

substitute let $z = e^{i\theta}$, then $d\theta = \frac{dz}{iz}$

and $\cos\theta = \frac{1}{2}\left(z + \frac{1}{z}\right)$ the limits 0 to 2π

becomes the unit circle $\underline{|z| = 1}$

$$I = \oint_{C,\ |z|=1} \frac{1}{\left(2 + \frac{1}{2}\left(z + \frac{1}{z}\right)\right) iz}\,dz = \oint_{|z|=1} \frac{2}{4z + z^2 + 1}\cdot\frac{dz}{i}$$

$$= \frac{2}{i} \oint_{|z|=1} \frac{1}{z^2+4z+1}\, dz$$

Find Poles

Solve $z^2+4z+1=0$
using quad. formula

$$z = \frac{-4 \pm \sqrt{16-4}}{2} = \frac{-4 \pm \sqrt{12}}{2} = -2 \pm \sqrt{3}$$

① $z_1 = -2 + \sqrt{3} \simeq -0.263$ (Res $|z|=1$)

② $z_2 = -2-\sqrt{3} \simeq -3.732$ (outside $|z|=1$)

Calculate Residue @ z_1

$$\text{Res}(z_1) = \lim_{z\to z_1} \frac{1}{z-z_1} = \frac{1}{(-2+\sqrt{3})-(-2-\sqrt{3})} = \frac{1}{2\sqrt{3}}$$

$$I = \frac{2}{i}\left[2\pi i \times \frac{1}{2\sqrt{3}}\right] = \frac{2\pi}{\sqrt{3}}$$

By Cauchy Residue theorem

$$\oint_C f(z)\, dz = 2\pi i\,[\text{Sum of residues}]$$

$$= 2\pi i\left[\frac{2}{i} \times \frac{1}{2\sqrt{3}}\right] = \frac{2\pi}{\sqrt{3}}$$

18/5/26